Experimental Intelligence in AI Scientists: From Initial World Models to Transferable Laws

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Abstract

Scientific agents never enter an experiment with a neutral mind: they carry an initial world model that may encode entities, mechanisms, parameter values and assumptions about how observations map to hidden states. A useful endpoint therefore does not show whether an agent discovered the governing relation, confirmed a correct model or succeeded despite a wrong one. We define initial-world-model intervention as a controlled variable in executable chemical worlds. The hidden world, public operations, evidence budget and safety surface are held fixed while the agent-facing model is changed at matched entity/ontology, structural/mechanistic, parametric or observation-model layers. The present completed development matrix and planned formal core instantiate the entity-level slice with opaque, aligned and misindexed dossiers; the other layers are registered extensions rather than post-hoc re-labellings of the same result. Each agent controls a multi-experiment campaign through one persistent operation-level session, records predictions and belief updates at fixed checkpoints, and commits an executable law summary and final recommendation. Evaluator-owned counterfactual queries, artifact-only transfer and blind replay separate evidence acquisition, law compression, action and generalization. In the completed DeepSeek development matrix, held-out prediction usually improved, but wrong-prior improvement did not exceed aligned-prior improvement (mean primary contrast -0.042 across 25 task-by-world clusters); executable law summaries often degraded the final explicit predictions, and no committed recommendation outperformed the observed incumbent. These observations support a focused research programme on experimental intelligence: how an agent turns an initial world model into evidence, an executable law, a decision and a transferable scientific artifact, and where that capability chain breaks.

1. Introduction

Scientific discovery is not simply the production of a high-scoring outcome. A researcher can obtain a useful result because an initial model was already correct, because evidence repaired an incorrect model, because a local heuristic happened to work, or because the endpoint was reached without a reusable account of the underlying relation. These explanations are scientifically distinct even when their final scores are identical. The corresponding object of study is therefore not a single prompt or material hint, but the agent's initial model of what exists in the world, how it works, which quantities matter and how measurements should be interpreted.

This distinction is increasingly important for AI systems that choose and execute experiments. Language-model agents can plan syntheses, call chemistry tools, operate instruments and participate in self-driving laboratory workflows [4, 5, 7, 15–17]. Interactive scientific environments likewise test whether agents can formulate hypotheses, acquire evidence and recover hidden rules [1, 8, 10, 12, 19, 20]. Yet an agent's apparent scientific success can remain difficult to interpret because pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization and verbal explanation are usually entangled.

The central problem is the initial world model. An agent entering an experimental campaign may carry assumptions about entity identities and properties, causal structure, parameter signs and ranges, or the reliability and meaning of observations. Any of these assumptions may be absent, useful or plausibly wrong. The important capability is not merely whether such information changes behavior. A scientifically adaptive agent should decide which evidence is worth acquiring, use observations to revise the relevant part of its model, reduce confidence in contradicted structure or parameters, summarize the recovered relation in an executable form and transfer that relation to conditions or compositions it has not directly tested. Conversely, an agent may preserve an incorrect model through selective measurement, reinterpret contradictory observations or patch only its action policy.

Physical self-driving laboratories are indispensable for real-material validation, but they make this causal question difficult to study at scale. Strictly matching hidden laws, material identities, measurement noise, budgets and safety conditions across alternative initial-model states is expensive and often impossible. ChemWorld provides a complementary experimental instrument: chemical entities, process structure, parameters, instrument mappings and private laws can be instantiated as executable worlds, while the public experimental interface and complete operation history remain controlled. This programmability permits the agent-facing initial model to change at a chosen layer while the external world and evidence opportunity remain fixed.

Here we introduce a controlled framework for studying experimental intelligence from initial world models to transferable laws. It makes four contributions.

1. **The initial world model becomes a layered intervention.** Entity/ontology, structural/mechanistic, parametric and observation-model assumptions can each be made opaque, aligned or deliberately wrong while the executable world remains fixed.
2. **Discovery is evaluated through evidence-conditioned transitions, not self-report alone.** Fixed checkpoints bind beliefs to evaluator-owned counterfactual queries and to the next experimental operation selected by the agent.
3. **Endpoint success is separated from reusable understanding.** Blind outcome replay, executable law summaries and context-reset artifact transfer distinguish local optimization from law recovery.
4. **The complete experimental process remains auditable.** One persistent session controls multiple experiments under a shared resource ledger, while failures, invalid actions, stopping and exact replay remain part of the outcome rather than being silently repaired.

Development results already establish a consequential boundary in the entity/ontology layer: explicit information reshapes behavior, but nominal-information warnings and endpoint gains do not reliably reveal whether the supplied mapping is correct. The formal core therefore asks whether autonomously acquired evidence selectively repairs that wrong mapping. Registered extensions ask whether the same capability holds when the error lies in causal structure, parameterization or observation semantics, and whether the resulting law survives transfer without the original session context.

2. Related work

2.1 Autonomous experimentation and laboratory agents

Autonomous chemistry systems combine planning, analysis and physical execution. Coscientist, ChemCrow, A-Lab and instrument-facing agents demonstrate tool use, synthesis planning, materials search and closed-loop experimentation on real equipment [4–7, 15, 16]. These systems establish that AI agents can participate in consequential experimental workflows. Their main evidential strength is physical validity; their practical constraint for the present question is the cost of repeatedly constructing matched alternative worlds in which only the correctness of a prior changes.

2.2 Virtual chemistry and process environments

Summit and Olympus support repeatable reaction optimization and experimental-design benchmarks, while PC-Gym exposes nonlinear process-control problems [3, 9, 11]. ChemGymRL provides modular interactive benches for reaction, extraction, distillation and characterization, enabling reinforcement-learning agents to act within a virtual chemistry laboratory [2]. MADE extends closed-loop discovery benchmarks to budget-constrained materials settings [13]. These systems make experimentation cheaper, safer and more repeatable than physical execution, but they are generally used to compare policies or optimize endpoints under one task definition. The present study instead uses world programmability to hold the executable task fixed while intervening on the agent's initial scientific model.

2.3 Interactive scientific-discovery benchmarks

DiscoveryWorld, BoxingGym, SciGym, SciExplorer and NewtonBench organize tasks around repeated cycles of hypothesis formation, intervention and inference [8, 10, 12, 14, 20]. CausaLab and ReplaySCM further emphasize interventional causal discovery and executable mechanism induction [1, 19]. SciDisco uses process-verifiable discovery environments to provide intermediate training signals [18]. These studies motivate evaluating the trajectory of discovery rather than only the final answer. Our focus is complementary: the hidden law does not change during a matched campaign; what changes is the entity, structural, parametric or observation-level model with which the agent enters that world.

2.4 The unresolved identification problem

Three ambiguities remain when endpoint score, belief statement and scientific understanding are not separated. First, a correct prior can make an agent look like a rapid discoverer even if it merely confirms supplied information. Second, a wrong prior can improve an endpoint by encouraging useful exploration without ever being rejected. Third, a verbal law summary can be correct while the agent's subsequent predictions or actions remain inconsistent with it. A matched prior intervention, typed checkpoints and evaluator-owned tests are needed to distinguish these cases.

3. Conceptual framework

3.1 Fixed world, programmable initial world model

Each matched comparison begins with one executable world containing a fixed evaluator-owned law. The public task, action space, actual observation channels, resource card, safety rules and bound stochastic identity are held constant. We intervene only on the agent-facing initial world model M_0: the compact set of assumptions available before the first experiment. This separation is essential. Changing the hidden law or public contract would create a different task; changing M_0 creates a controlled epistemic intervention within the same task.

The programmable intervention space has four scientific layers and one non-intervention boundary.

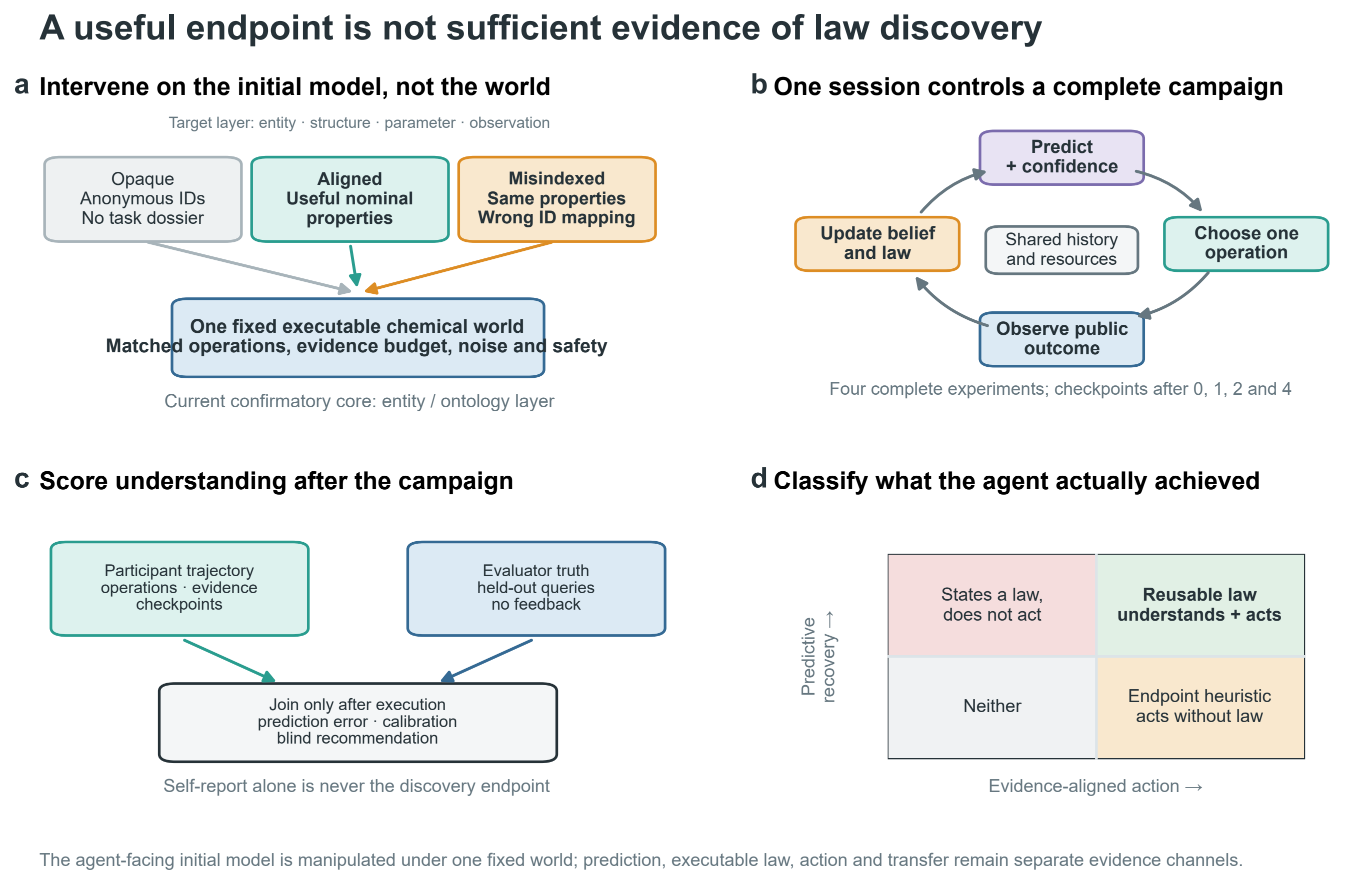

Figure 1. From an initial world model to a reusable law. a, The current entity-level instantiation uses opaque, aligned and misindexed dossiers in the same fixed executable world; the same intervention logic can target structural, parametric or observation-model assumptions. **b,** One persistent session repeatedly predicts, selects an operation, observes the public outcome and updates its belief and executable law summary across a shared-resource campaign. **c,** Participant trajectories and evaluator-owned held-out truth remain separate until the campaign reaches a terminal state; prediction error, calibration and blind recommendation outcomes are scored afterward. **d,** Predictive recovery and evidence-aligned action define four distinguishable phenotypes. Only their joint success, followed by transfer, supports a reusable-law claim; endpoint success or a correct statement alone does not.

Table 1. Programmable layers of the agent's initial world model. The external executable world and public contract remain fixed within every matched comparison.

Initial-model layer	What may be aligned or wrong	Role in this paper
Entity / ontology	identity–property mappings, entity classes and property bundles	current development evidence and confirmatory core
Structural / mechanistic	causal topology, active process modules and dominant-pathway assumptions	separately registered extension
Parametric	coefficient signs, thresholds, orderings and plausible ranges	separately registered extension
Observation model	instrument mapping, reliability, bias and noise assumptions	secondary diagnostic probe
Contract / resource boundary	budget, safety, action permissions and actual observation interface	authoritative and fixed; never treated as a manipulable prior

Within any selected layer, intervention quality follows the same logic:

- **Opaque:** no additional task-specific claim is supplied for the manipulated layer.
- **Aligned:** incomplete information is directionally consistent with the fixed world.
- **Misspecified:** an equally detailed and equally confident model is wrong at the frozen target layer.

The current entity/ontology implementation realizes the misspecified condition through a misindexed dossier: the same property bundles, fields, values, wording and confidence language are retained, but the bundles are permuted across material identifiers. Structural, parametric and observation-model interventions require their own matched encodings and identifiability qualification; they cannot be declared equivalent to material misindexing after observing the result. Experimental evidence remains explicitly authoritative in all conditions. A wrong initial model is therefore not a trick question about obedience; it tests whether the agent seeks and uses evidence that can override a plausible scientific representation.

3.2 Operation, experiment and campaign

An operation is one submitted physical or measurement action. A complete experiment begins with a new batch and ends with a committed final assay or an allowed discard. A discovery campaign comprises multiple complete experiments under one fixed world, one persistent agent session and one shared resource ledger. Physical batch state resets between experiments, but the public history, agent context, hidden law and remaining resources persist.

This distinction matters because repeated operations and experiments inside one campaign are nested observations, not independent scientific samples. The independent analysis unit is the matched task-by-world cluster.

3.3 Four separable outcomes and one capability chain

We distinguish four outcomes that are often collapsed into a single score, while recording the intermediate transitions that connect them.

1. **Endpoint optimization:** whether the campaign identifies a high-quality experimental outcome.
2. **Predictive recovery:** whether held-out counterfactual prediction error decreases.
3. **Prior correction:** whether evidence selectively improves the wrong-prior condition without degrading the correct-prior condition.
4. **Reusable law recovery:** whether the final executable summary predicts unseen continuous conditions and supports held-out control or transfer.

The process-level chain is therefore

$$\text{initial world model} \rightarrow \text{experiment selection} \rightarrow \text{evidence acquisition} \rightarrow \text{prediction / belief update} \rightarrow \text{executable law} \rightarrow \text{action} \rightarrow \text{transfer}$$

The paper reports transition losses rather than a composite intelligence score: prediction-to-law loss, law-to-action inconsistency and action-to-transfer loss. This makes it possible to identify where a capability fails without treating a successful endpoint as proof that every upstream step was correct.

An agent can succeed on any subset. In particular, endpoint success without predictive and transfer validity is classified as local optimization rather than law discovery.

4. Study design

4.1 Chemical-world cohort and intervention studies

The formal public cohort spans five task families: electrochemical conversion, reaction followed by crystallization, reaction followed by distillation, phase-partition discovery and safety-constrained reaction. Five independently selected public worlds per task yield 25 task-by-world clusters. Each cluster contains the three entity/ontology arms, for 75 participant cells. Development and method-qualification worlds are disjoint from this cohort. A separate private cohort is committed in advance and remains sealed until public analysis is frozen.

This 75-cell matrix is the primary, entity-level slice of a broader initial-world-model programme; it is not a claim that material dossiers exhaust the space of scientific priors. The studies are organized so that breadth does not become an uncontrolled benchmark:

1. **Study A — Prior-conditioned free discovery.** The current five-task, three-arm matrix tests whether an agent acquires evidence, updates predictions and rejects a wrong entity/ontology model under a fixed world and shared campaign ledger.
2. **Study B — Matched-evidence falsification.** A cloned-world secondary probe presents the same contradictory evidence to each arm, separating failure to seek evidence from failure to update after seeing it. It uses independent sessions and is excluded from Study A denominators.
3. **Study C — Executable law and action.** Typed law summaries, held-out predictions and blind recommendations test the transitions from prediction to law and from law to action; no verbal statement alone counts as discovery.
4. **Study D — Artifact-only compositional transfer.** After source-world learning, raw evidence, prose summaries or executable laws are transferred to a context-reset agent in a new combination. No-artifact, trajectory and typed-law conditions are compared, and within-family replication is kept separate from genuine compositional transfer.

Structural/mechanistic, parametric and observation-model interventions are registered extensions to Study A rather than hidden post-hoc subgroups. Each extension must first pass its own identifiability and matched-information preflight; the experiment matrix is intentionally sparse and selected by mechanistic coverage, not expanded into a full factorial by default.

Environment qualification and participant outcomes form separate evidence layers. Environment tests establish that the hidden relations are coherent, identifiable and executable through the public measurement surface. They do not show that an agent discovers those relations.

4.2 Persistent experimental agent (Study A)

Each cell is controlled by one persistent Codex process and one provider session. After every public outcome, the participant chooses the next operation through the host-owned laboratory tool. The host validates schemas, executes transactions, updates resources and protects private state, but does not select or repair scientific actions.

Each campaign contains four complete experiments and typed checkpoints before evidence, after the first and second experiments, and at the end of the fourth experiment. A checkpoint records the agent's assessment of initial-model reliability at the manipulated layer, predictions, uncertainty, evidence references, executable law summary

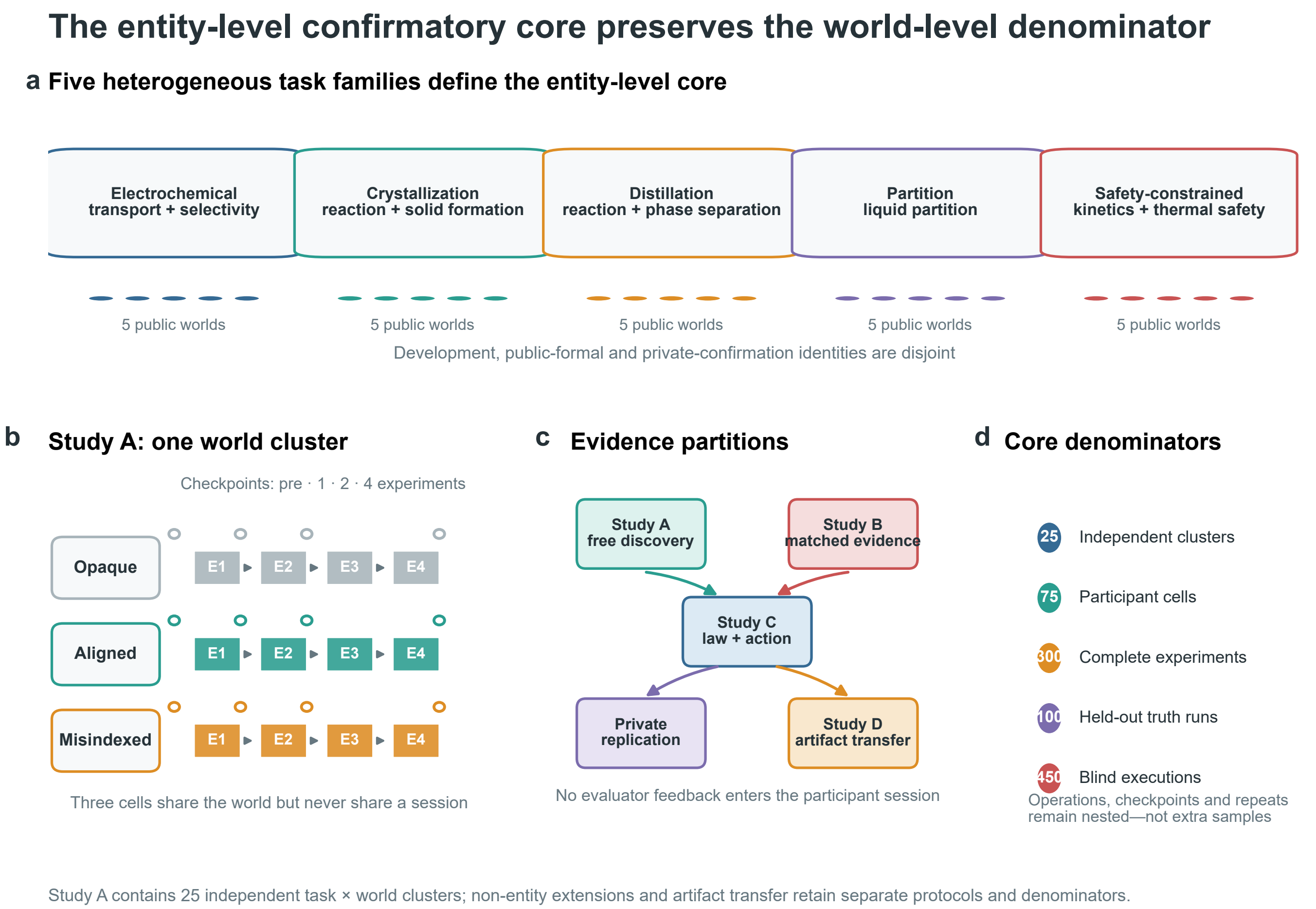

Figure 2. Entity-level confirmatory core and separated study architecture. a, Five task families each contribute five public-formal worlds to Study A, selected independently of participant outcomes; development, public and private identities are disjoint. **b,** Each independent task–world cluster contains opaque, aligned and misindexed entity/ontology cells. Every cell uses one persistent session for four complete experiments, with checkpoints before evidence and after one, two and four experiments. **c,** Free discovery, matched-evidence falsification, evaluator truth and action tests, private within-family replication and artifact-only compositional transfer retain separate sessions, resources and denominators. **d,** The planned public denominator comprises 25 independent clusters, 75 participant cells and 300 complete experiments, with 100 provider-free held-out truth executions and 450 provider-free blind executions nested within the corresponding clusters. These are frozen design denominators, not completed outcomes.

and next experimental intent. Checkpoints do not create additional provider sessions.

4.3 Evaluator-owned evidence

Four registered counterfactual queries per task-world cluster are executed independently by the evaluator. Their truth is shared across prior arms and checkpoints and is never returned to the participant. The primary prediction error is the mean normalized absolute error across registered query-metric pairs.

The evaluator also executes each final typed law summary on the same registered query coordinates. This produces a separate cell-level record of schema validity, complete query-metric executability, truth-normalized error, error change relative to the pre-evidence and effective-final checkpoints, and consistency with the participant's final explicit predictions. These public measures are descriptive: no post-hoc binary validity threshold is applied, and reusable-law status remains unavailable without the prespecified private transfer test.

After the campaign, the participant commits one completed experiment as its final recommendation. The evaluator then performs paired blind replay of the observed incumbent and the committed recommendation. These executions use separate resources and do not enter participant-operation or provider denominators.

4.4 Hypotheses and estimands

Let $E_{a,k}$ denote held-out prediction error for prior arm a at checkpoint k. The primary hypothesis tests selective evidence-driven correction:

$$C_{\text{prior}} = (E_{\text{misindexed,pre}} - E_{\text{misindexed,final}}) - (E_{\text{aligned,pre}} - E_{\text{aligned,final}}).$$

Success requires the lower confidence bound for C_{prior} to exceed zero, the wrong-prior condition to improve, and the aligned condition not to deteriorate beyond a prespecified tolerance. Correct-prior utility, wrong-prior vulnerability and knowledge-to-action translation form a hierarchical secondary family. For the structural, parametric and observation-model extensions, the same estimands are re-bound to the manipulated layer before execution; their results cannot be pooled with the entity-level primary contrast unless the intervention semantics and denominators are identical. Endpoint, calibration, behavior, law-summary, transfer, resource and safety outcomes are reported as separate channels rather than one leaderboard score.

Failed scientific cells remain in the denominator and are not replaced. A right-censored cell carries its last valid checkpoint forward; a missing final prediction receives zero primary improvement. Only a pure infrastructure failure without persisted trajectory may resume under the frozen attempt cap.

5. Development evidence

The following results qualify the method and sharpen the scientific question. They are not part of the public formal or private-confirmation denominator, and the two provider configurations are not used for a cross-provider capability ranking. They instantiate only the entity/ontology layer of the initial-world-model intervention matrix; they should not be read as evidence that structural, parametric or observation-model priors have already been tested.

5.1 Entity-level priors reshape endpoint behavior

One development matrix used the frozen persistent-session interface with a WellAU-provided Codex model. It produced 44 completed cells out of 45 and 176 complete experiments out of 180. Mean paired aligned-minus-opaque differences in the best observed endpoint were +0.211 for electrochemical conversion, +0.057 for crystallization and -0.036 for distillation, with the distillation contrast based on four complete pairs. Misindexed information was not consistently harmful, and explicit misindex warnings included substantial false positives in aligned cells.

A recovery-amended DeepSeek development block was completed across all five task families, retaining the immutable seed-0 gate records for partition discovery and safety-constrained reaction and adding their seeds 1–4 in a separate continuation block. No seed-0 outcome was rerun or replaced. Every scheduled cell reached a terminal record (**75/75**); **69/75** cells were completed and qualified, with **290/300** complete experiments, **2,663/2,587** operation attempts/committed operations, **73** validation failures, **3** resource rejections, **69** recovered MCP failures and **0** provider-error events. Exact physical/resource replay passed for **75/75** terminal trajectories. Provider accounting covered 72/75 cells; the remaining three stopped before a provider terminal event and retain usage as unavailable rather than zero. The complete development block is still descriptive: it is not part of the preregistered public matrix, has no private transfer confirmation or formal hypothesis test, and provider groups are never pooled into a capability ranking.

The ordering is not aligned, opaque, then misindexed. In distillation, every paired seed favored both explicit-information arms over opaque identifiers, and the misindexed mean gain was larger. In partition discovery, both explicit-information contrasts were negative, whereas the safety task was mixed. These task-pattern differences are descriptive and are partly coupled to the continuation contract; they do not support a pooled provider effect. A better endpoint therefore cannot be treated as evidence that the agent accepted a correct prior, rejected a wrong prior or recovered the hidden law.

5.2 Held-out prediction improves, but entity-level correction is not selective

The post-hoc development evaluator executed four registered counterfactual queries for each of the 25 task-by-world clusters. All **100/100** truth queries completed with exact replay and without a provider call. Final checkpoint predictions were scoreable for **72/75** retained participant cells. Using the frozen failure rules, aligned-prior prediction error improved in 24/25 cells and misindexed-prior error improved in 22/25 cells. Improvement was therefore common, but it was not selectively stronger under the wrong prior: the primary contrast had a descriptive mean of -0.042, median -0.039, and was positive in only **7/25** clusters. Restricting description to the 19 complete-case clusters gave the same direction (mean -0.039; 6/19 positive). Only the safety-constrained task had a positive task-level mean; the other four task means were negative.

The evaluator executed **71/75** final typed law summaries on the same registered coordinates. The summary improved on the final explicit predictions in 12/23 opaque, 7/24 aligned and only 3/24 misindexed cells; it was worse in the remaining 11, 17 and 21 cells, respectively. Thus, an agent can improve its checkpoint predictions without compressing that improvement into a reusable executable law. Blind replay sharpened the action boundary: **414/414** scheduled executions completed across 69 qualified cells, but the committed recommendation beat the observed incumbent in 0 cells, was equivalent in 66 and was worse in 3. These results do not prove that correction is impossible; they show that endpoint gains, prediction repair, law compression and recommendation quality are distinct development outcomes.

5.3 Verbal suspicion is not selective correction

Across available terminal belief records, final DeepSeek misindex warnings were 0/5, 5/5 and 3/5 for opaque, aligned and misindexed electrochemical cells; 0/4, 5/5 and 4/4 in crystallization; 0/5, 5/5 and 5/5 in distillation; 0/4, 2/4 and 3/4 in partition discovery; and 0/5, 2/4 and 0/5 in safety-constrained reaction. The model therefore often associated dossier presence with possible misindexing, but the pattern was task-dependent and did not selectively distinguish the correct and incorrect dossiers. Mean aligned-minus-misindexed changes in self-reported prior reliability were small or heterogeneous across tasks, including negative changes in several continuation cells.

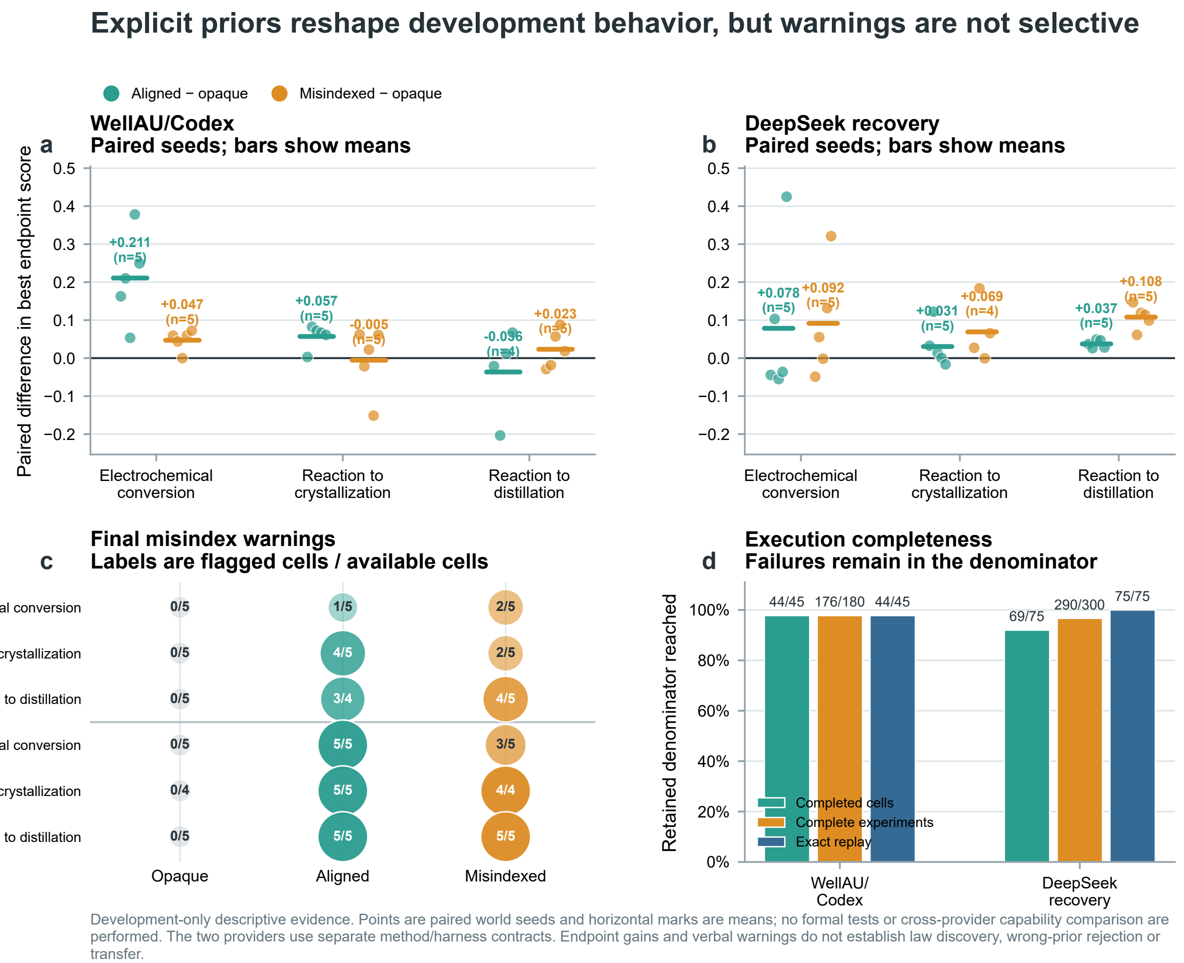

Figure 3. Provider-separated development evidence for prior-sensitive behavior. a,b, Paired world-seed differences in the best endpoint observed during four-experiment campaigns for aligned versus opaque and misindexed versus opaque information. Points are retained paired seeds and horizontal bars are descriptive means. WellAU/Codex contains five pairs per task except the aligned distillation contrast ($n = 4$); DeepSeek recovery contains five pairs except the misindexed crystallization contrast ($n = 4$). **c,** Final explicit misindex warnings, shown as flagged cells over available terminal belief records for every provider, task and prior arm. **d,** Completed-cell, complete-experiment and exact-replay denominators. Failures remain in the scheduled or terminal denominator and are not replaced. Panels a–c retain the common three-task paired endpoint/warning source used for provider-separated continuity; the complete five-task DeepSeek operational denominator is shown in the closeout table. All panels are development-only descriptive summaries; no confidence interval, formal hypothesis test or cross-provider capability comparison is performed. Endpoint gains and verbal warnings do not establish law discovery, selective wrong-prior correction or transfer.

Table 2. Five-task DeepSeek development closeout. All five task families reached the scheduled five-seed terminal denominator. The partition and safety continuation preserved their immutable seed-0 gate outcomes; their operational rows are reported descriptively and are not pooled with the common three-task paired endpoint panels.

Task	Terminal	Qualified	Experiments	Attempts/committed	MCP failures	Exact replay
Electrochemical conversion	15/15	15/15	60/60	373/372	13	15/15
Reaction to crystallization	15/15	13/15	54/60	606/605	18	15/15
Reaction to distillation	15/15	15/15	60/60	637/637	7	15/15
Partition discovery	15/15	12/15	58/60	553/501	14	15/15
Safety-constrained reaction	15/15	14/15	58/60	494/472	17	15/15

This is a scientifically useful negative boundary. A warning token or reduced stated confidence is not a valid bias-rejection endpoint unless it predicts evaluator-scored correction and subsequent evidence-aligned action.

5.4 Persistent-session accounting exposes a separate operational layer

The 75 terminal DeepSeek trajectories accumulated **267,929,149 input tokens**, including **260,033,536 cached tokens** (97.05%), **7,895,613 uncached input tokens** and **2,932,468 output tokens**. These are cumulative turn-level provider counters for long-lived sessions. The large cached fraction therefore reflects reuse of the

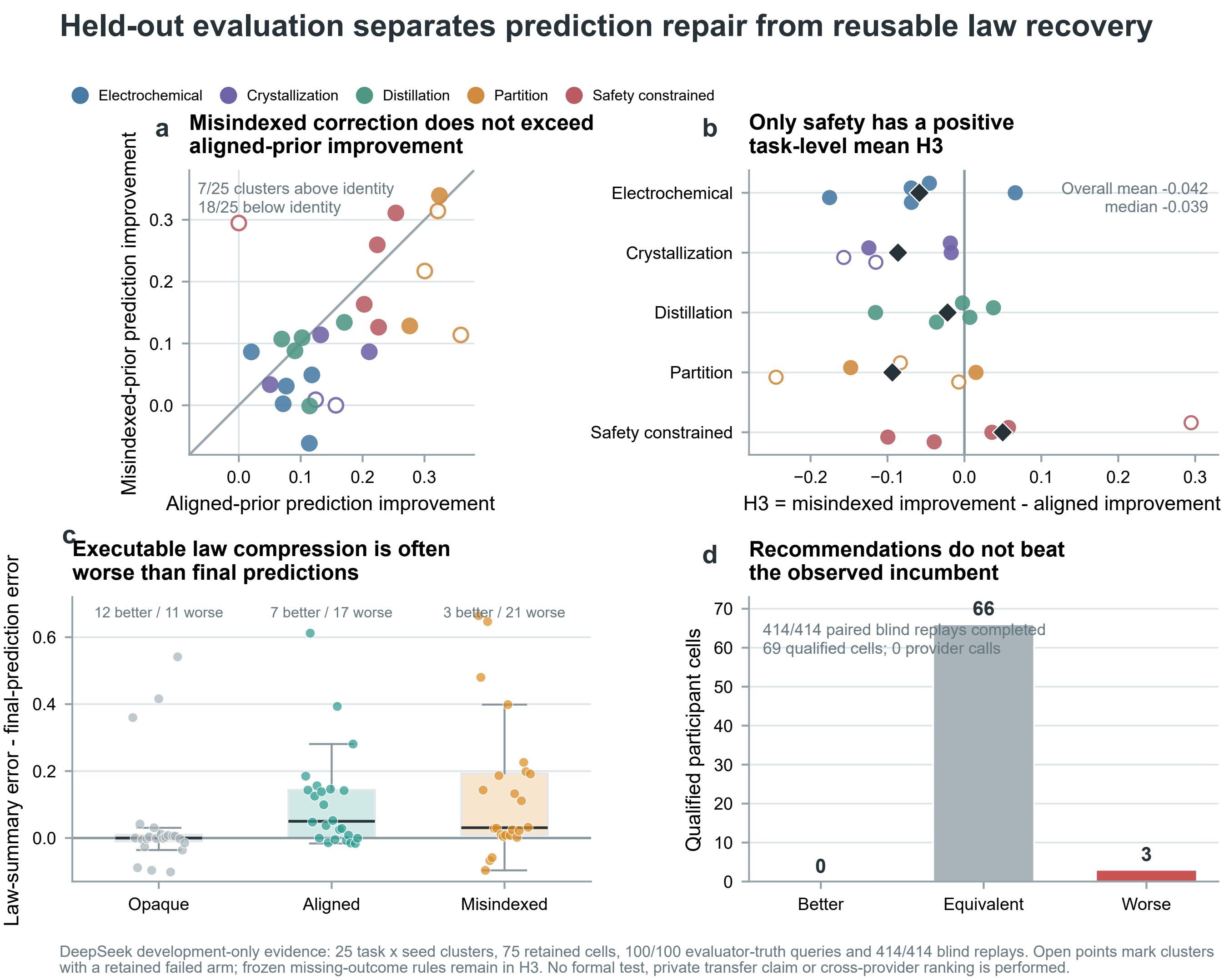

Figure 4. Held-out evaluator confirmation of the five-task DeepSeek development matrix. a, Aligned-prior and misindexed-prior reductions in normalized held-out prediction error for all 25 task-by-world clusters. The identity line marks equal improvement; filled points are complete three-arm clusters and open points retain at least one failed arm under the frozen missing-outcome rule. **b,** The primary development contrast C_{prior} by task and seed. Diamonds are task means; positive values favor greater correction in the misindexed arm. **c,** Executable law-summary error minus final explicit-prediction error for 71 evaluable summaries. Negative values indicate beneficial compression. **d,** Paired blind replay of the committed recommendation versus the observed incumbent for 69 qualified cells. The evaluator completed 414/414 replays with zero provider calls. All panels are development-only descriptive evidence; no formal test, private transfer claim or cross-provider ranking is performed.

Table 3. DeepSeek five-task development endpoint contrasts. Values are descriptive paired-seed means in best endpoint score; the partition and safety rows are continuation observations and are not pooled into the common three-task Figure 3 endpoint panels.

Task	Aligned–opaque	Misindexed–opaque
Electrochemical	$+0.0785$ ($n=5$)	$+0.0915$ ($n=5$)
Crystallization	$+0.0305$ ($n=5$)	$+0.0690$ ($n=4$)
Distillation	$+0.0374$ ($n=5$)	$+0.1080$ ($n=5$)
Partition	-0.1397 ($n=5$)	-0.1063 ($n=5$)
Safety-constrained reaction	$+0.0223$ ($n=5$)	-0.0196 ($n=5$)

shared prompt and growing campaign history; it is not repeated model output and does not represent additional independent experimental evidence. Usage was reconciled for 72/75 cells. The other three cells stopped before a provider terminal event, so their usage remains unavailable rather than being imputed as zero.

Operational failures also require their own denominator. The matrix recorded 73 schema-validation failures and 69 recovered MCP tool failures but no provider-error events. Moreover, 76 of 2,663 operation attempts did not become committed operations. Thus, a zero provider-error count would have hidden most of the execution burden: model-to-tool conformance, recovery and resource preflight were more consequential than provider transport in this block. These events are properties of the complete agent system and remain separate from both physical outcomes and independent world-level scientific samples.

5.5 Development conclusion

Across the retained development configurations, explicit priors clearly alter the course and endpoint of experimentation, and most cells improve their held-out predictions. The same evidence does not show selective wrong-prior rejection: aligned improvement exceeds misindexed improvement on average, typed law compression is frequently lossy and committed recommendations do not outperform the observed incumbent. The formal study is therefore necessary for prospective inference, while the development block already establishes that prediction repair, law recovery and action quality must remain separate outcomes.

6. Formal results

This section will report the preregistered public matrix only after every scheduled cell has reached a completed, right-censored or failed terminal state. Arm contrasts will remain blinded during execution. The section is reserved for:

- the primary selective-correction contrast and its three required conditions;
- correct-prior utility and wrong-prior vulnerability;
- checkpoint-level prediction-error and calibration trajectories;
- executable law-summary coverage, continuous counterfactual error, compression stability and consistency with final explicit predictions;
- evidence-seeking, selective measurement, attribution and prior-persistence phenotypes;
- blind incumbent-versus-recommendation outcomes;
- complete failure, censoring, resource, token, cost and wall-time denominators.

The primary formal result is the entity/ontology intervention in Study A. Any structural, parametric or observation-model extension will be reported in a separately identified subsection with its own intervention audit and denominator; it will not be silently merged into the primary arm contrast.

No development value will be substituted for a missing formal result.

7. Private confirmation and transfer

After the public method and analysis code are frozen, the committed private cohort will be executed once. The same prediction, law-summary and control metrics will test world-held-out transfer within the five registered task families. Private identities will not be revealed to the participant, and a negative or incomplete result will not trigger a replacement run. This within-family replication is not compositional transfer. Study D requires a context reset and an artifact-only hand-off into a new task combination, with no-artifact and raw-trajectory controls; a mechanism-family-held-out result would require a separately registered extension.

The strongest claim—discovery of a reusable law—requires joint evidence: selective wrong-prior correction, aligned-prior non-degradation, an executable typed law summary, valid predictions on unseen continuous conditions and successful preregistered transfer. Anything less will be reported as confirmation, local correction, policy repair or endpoint optimization according to the observed combination.

8. Discussion

8.1 Why endpoint success is insufficient

The development matrices show that a deliberately wrong dossier can improve the best observed endpoint. This is not paradoxical. A prior can change where an agent searches, increase experimental diversity or direct attention toward a useful region even when its material mapping is wrong. Endpoint improvement measures utility of the induced search trajectory, not truth of the agent's scientific model.

8.2 Bias rejection must be behaviorally selective

The broad tendency to distrust any explicit dossier illustrates a second ambiguity. Generic skepticism can generate the right verbal stance without identifying the wrong prior. A valid correction measure must be selective: contradictory evidence should improve wrong-prior predictions more than correct-prior predictions, and that improvement should influence subsequent experimental choices.

8.3 Scientific understanding as a coupled capability

Prediction, explanation and action can dissociate. An agent may understand a relation but lack the operational competence or remaining resources to exploit it; it may act successfully without an accurate model; or it may state the right model while continuing to choose inconsistent experiments. The joint evaluation therefore supports four interpretable phenotypes: understands and acts, understands but cannot act, acts without understanding, and neither.

8.4 The harness is part of the evaluated agent system

A persistent Codex session contributes capabilities that a stateless model call does not: it retains the campaign history, chooses among tools after each observation and can revise a plan without a host reconstructing its reasoning state. The same harness also introduces failure surfaces through tool discovery, schema conformance, retry rules, context growth and checkpoint submission. Provider-side caching changes resource use but not the number of experiments or independent worlds. Consequently, the participant is the frozen combination of model, reasoning setting, prompt, Codex runtime, MCP interface and resource policy—not the model weights in isolation. A cross-model claim requires these components to be matched or explicitly manipulated; the present development providers are therefore reported separately.

8.5 Scope and limitations

The study evaluates bounded executable chemical worlds rather than universal chemical fidelity or direct wet-laboratory validity. A single frozen participant method supports conclusions about that agent-system configuration, not language models in general. Five task families provide mechanistic breadth but only five independent public worlds per task, so the confirmatory design is powered for moderate-to-large effects rather than subtle differences. Evaluator queries are registered before participant execution, but any finite query set samples only part of a hidden law. The private cohort tests new worlds within the same task families rather than transfer to an unseen mechanism family. The current completed data also manipulate only the entity/ontology layer; structural, parametric and observation-model interventions remain registered extensions until their independent preflights and formal blocks are run. Finally, exact software replay does not eliminate model-provider variability; provider attempts and session failures are reported as operational characteristics rather than independent scientific samples.

9. Methods

9.1 World and initial-model construction

Each task instantiates a public experimental contract and a private evaluator-owned material or process law. Public formal worlds are selected deterministically from a namespace disjoint from development worlds. Within each world cluster, the three entity/ontology arms share the hidden law, public contract, resource card and stochastic identity. Aligned and misindexed dossiers contain identical fields, values, wording and confidence language; the latter applies a frozen permutation to the material identifiers. Structural, parametric and observation-model extensions alter only their declared agent-facing representation while retaining the same external world and contract. A layer extension is admitted only after a separate identifiability audit confirms that its aligned and misspecified encodings are matched in information volume, wording and confidence.

9.2 Transactional execution and resources

Every operation enters schema validation and resource preflight before candidate execution. Committed operations update physical state and the campaign ledger; invalid or resource-rejected attempts retain their declared reporting debit without entering committed physical state. Task-specific resource cards bound vessel starts, assays, measurements, stocks, process time, repeated operations, quench and transfer time, and closeout reserve across all experiments in a campaign.

9.3 Persistent Codex/MCP execution

Each participant cell launches one Codex Responses process and retains one provider session across the four complete experiments. Web search is disabled. The participant instructions prohibit shell use, file changes and repository inspection and require physical decisions to pass through the host-owned `chemworld_lab` STDIO MCP server. The bounded domain tools expose material information, belief checkpoints, operation submission, public status and history, artifact inspection and final recommendation commitment. The host validates and executes submitted actions but never chooses a fallback scientific action.

Every operation submission contains a bounded decision audit stating its expected effect, diagnostic target and evidence dependence. Tool receipts retain call order, status, timestamps, error classes and argument/result hashes without retaining raw provider payloads or private chain-of-thought. A provider retry or infrastructure resume is an operational attempt within the same cell, not a new experiment or independent sample. Belief checkpoints are MCP calls inside the existing session rather than separate provider conversations.

9.4 Belief and law-summary checkpoints

Checkpoint records contain prior assessment, predictions, uncertainty, evidence references, an executable law summary, next-experiment intent and overall confidence. The schema permits bounded rationales but not an unconstrained persistent notebook. After the campaign, explicit predictions and the final law summary are evaluated against sealed truth packs. The summary must execute for the exact registered query-metric set; the analysis records its normalized error, pre-to-summary improvement, error relative to the effective final checkpoint and prediction-consistency error. These quantities are not converted into a public binary law-discovery decision before private transfer.

9.5 Statistical analysis

The independent units are the 25 task-by-world clusters. The primary contrast is estimated with task fixed effects and a one-sided 0.05 criterion under an intersection–union rule. Three secondary hypotheses use Holm familywise correction. Task-specific contrasts are reported as heterogeneity estimates rather than independent confirmatory claims. Prespecified sensitivity analyses include complete-case, worst-case failed-arm, heteroscedasticity-robust and task-stratified cluster-bootstrap analyses.

9.6 Private confirmation boundary

Private confirmation is derived only after the public 75-cell matrix is terminal and the public confirmatory analysis, participant method and analysis source are hash-bound. An external seal commits five new world identities for each of the five registered task families. The validator checks that the 25 identities are unique, lie inside the private namespace and are disjoint from development and public-formal worlds. The repository records only the commitment; the seal nonce and identity schedule remain outside version control.

The sealed schedule deterministically expands to 75 prior-arm cells with the same participant, campaign-resource, checkpoint, prediction, executable-summary and blind-replay contracts used in the public matrix. A preflight cannot authorize execution: it remains blocked until a separate private currency ceiling, one-shot command approval and final runner/analysis release receipt exist. Its identity-bearing artifact can be created only once under the ignored private run root. A completed or failed private cell is never replaced because of its result; at most one missing-infrastructure-only resume is permitted per cell, and all failures and unstarted cells remain in the denominator.

9.7 Reproducibility and failure accounting

Participant trajectories, evaluator truth packs and blind-replay packs are stored separately and joined through immutable receipts. Every terminal participant trajectory must pass physical replay, campaign-resource replay and hidden-boundary audit. Provider process attempts, provider sessions, tool calls, operation attempts, committed operations, complete experiments, cells and evaluator executions are reported with distinct denominators.

10. Data and code availability

The executable environment, frozen protocols, analysis code, source data and reproducible figure scripts will accompany the public release. Raw provider payloads, credentials and private world identities are excluded. The private cohort commitment will be released before execution, and the identities and results will be disclosed with the final confirmation package.

11. Author contributions

Jiangjie Qiu, Yijun Li and Yaotian Yang contributed equally. A role-based contribution statement for all six authors will be finalized before submission without changing the author order or corresponding-author designation.

12. Competing interests

The authors declare no competing interests.

References

[1] Serafim Batzoglou. ReplaySCM: A benchmark for executable causal mechanism induction from interventions, 2026. Preprint.

[2] Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3:742–758, 2024. doi: 10.1039/D3DD00183K.

[3] Maximilian Bloor, José Torraca, Ilya Orson Sandoval, et al. PC-Gym: Benchmark environments for process control problems, 2024. Preprint, arXiv:2410.22093.

[4] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

[5] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.

[6] Tianyi Dai, Sriram Vijayakrishnan, Filip T. Szczypiński, et al. Autonomous mobile robots for exploratory synthetic chemistry. *Nature*, 635:890–897, 2024. doi: 10.1038/s41586-024-08173-7.

[7] Kourosh Darvish, Marta Skreta, Yuchi Zhao, et al. ORGANA: A robotic assistant for automated chemistry experimentation and characterization. *Matter*, 8(2), 2025. doi: 10.1016/j.matt.2024.10.015.

[8] Haonan Duan, Stephen Zhewen Lu, Caitlin Fiona Harrigan, Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, and Chris J. Maddison. Measuring scientific capabilities of language models with a systems biology dry lab, 2025. Preprint, arXiv:2507.02083.

[9] Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry–Methods*, 1:116–122, 2021. doi: 10.1002/cmtd.202000051.

[10] Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.

[11] Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.

[12] Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.

[13] Shreshth A. Malik, Tiarnan Doherty, Panagiotis Tigas, Muhammed Razzak, Stephen J. Roberts, Aron Walsh, and Yarin Gal. MADE: Benchmark environments for closed-loop materials discovery. In *International Conference on Machine Learning*, 2026.

[14] Maximilian Nägele and Florian Marquardt. Agentic exploration of physics models. *Physical Review X*, 16:031002, 2026. doi: 10.1103/xnqc-q6nt.

[15] Tao Song, Man Luo, Xiaolong Zhang, et al. A multiagent-driven robotic AI chemist enabling autonomous chemical research on demand. *Journal of the American Chemical Society*, 147(15):12534–12545, 2025. doi: 10.1021/jacs.4c17738.

[16] Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.

[17] Aikaterini Vriza, Michael H. Prince, Tao Zhou, Henry Chan, and Mathew J. Cherukara. Operating advanced scientific instruments with AI agents that learn on the job. *npj Computational Materials*, 12:160, 2026. doi: 10.1038/s41524-026-02005-0.

[18] Yucheng Xu, Keyi Zhang, Yuyang Yu, Min Zhang, Shiyuan Meng, Pei Chu, and Zhongying Tu. Scaling scientific discovery environments for turn-level agentic RL, 2026. Preprint, arXiv:2607.28990.

[19] Junlin Yang, Dylan Zhang, Xiangchen Song, et al. CausaLab: A scalable environment for interactive causal discovery toward ai scientists, 2026. Preprint.

[20] Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, et al. NewtonBench: Benchmarking generalizable scientific law discovery in llm agents. In *International Conference on Learning Representations*, 2026.